Pre-analysis plan #ManyDaughters

H1. ***Having daughters leads to stronger support for gender equality (as measured by attitudes).***

(X for respondents with daughters) > (X for respondents without daughters).

This study investigates whether having daughters influences individuals' support for gender equality. The analysis will use survey data from the 2019 wave of the SOEP.

Variables

Outcome variable: gender attitudes index

- The dependent variable is an index measuring support for gender equality, based on six survey items (plh0395i01 to plh0395i06) that capture attitudes toward gender roles. These variables are on the attitudinal items in the "Family Attitudes" module of the NSLY's "Attitudes and Expectations" section. The answers are coded in a seven-point scales where 1 represents an support of traditional gender roles (e.g., "strongly disagree" that it should be accepted in our society that not everyone identifies with their biological sex) and 7 represents a rejection of traditional gender roles (e.g., "strongly agree" that It's good that legislators take into account that not everyone can be classified as either male or female).
- A factor analysis and Cronbach's alpha will be used to assess whether the items form a single, reliable scale. If the items do not load onto a single factor, factor scores will be used instead. If they do, the index will be constructed by summing the six items, assuming they all load positively on the same dimension.

Main independent variable: has daughter

- A binary variable indicating whether the respondent has at least one daughter. It is coded as 1 if any of the child sex variables (kidsex01 to kidsex19) equals 2 (female), and 0 otherwise.

Instrumental variable: first child is a daughter

- To address endogeneity concerns, the sex of the first child is used as an instrument for having a daughter. This variable is coded as 1 if the first child is a girl (kidsex01 = 2), and 0 if the first child is a boy (kidsex01 = 1). The assumption is that the sex of the first child is randomly assigned and thus exogenous to the respondent's gender attitudes.

Control variables

- Age: calculated as the difference between the survey year and the respondent's year of birth.
- Sex: included as a categorical variable to control for gender differences in attitudes.
- Education: recoded from the ISCED classification (pgisced97) into five categories:
    - Low (levels 0–2)
    - Middle vocational (level 3)
    - Vocational + Abitur (level 4)
    - Higher vocational (level 5)
    - Higher academic (level 6)

- Household income: the natural logarithm of household net income (hghinc) is used to reduce skewness. Only non-negative values will be retained.

All variables will be weighted using the appropriate survey weights (phrf) to ensure representativeness.

Inclusion and exclusion criteria

- Survey year: only data from the 2019 wave of the SOEP will be used, as the questions on attitudes were only available in that year.
- Age range: individuals from the age 18 and 60 years old will the included in the sample to capture working-age adults.
- Parental status: only individuals with at least one biological child will be included, as the analysis focuses on the effect of having daughters, and to apply the instrument that condition is necessary.
- Missing Data: Respondents with missing or invalid values in key variables (e.g., gender attitudes items or child sex) will be excluded from the final analysis.

Exact analysis

To test the hypothesis that having daughters leads to stronger support for gender equality, I will estimate a two-stage least squares (2SLS) instrumental variable regression. This method is essential for identifying causal effects, as it addresses potential endogeneity in the relationship between having daughters and gender attitudes. An instrument should be a variable that affects whether someone has a daughter but does not directly affect gender attitudes. The variable chosen in the sex of the first child since whether the first child is a girl is random. The strength and validity of the instrument will be assessed. The main regression model is specified as follows:

- Dependent variable: genderatt – index of gender attitudes.
- Endogenous independent variable: hasdaughter – dummy for having at least one daughter.
- Instrumental variable: first_daughter – dummy for whether the first child is a girl.
- Controls: educ – categorical variable for education level, age , sex and log_hhnetincome – log of household net income.

Interpretation of results

To evaluate the hypothesis, I will focus on the estimated coefficient of the variable hasdaughter in the second stage of the instrumental variable regression. The coefficient indicates the direction and magnitude of the effect and the p-value assess statistical significance.

If the coefficient on hasdaughter is positive and statistically significant, this will be interpreted as supporting the hypothesis that having daughters leads to more egalitarian gender attitudes.

If the coefficient is not statistically significant, or if it is negative, the result will be interpreted as not supporting the hypothesis.

H2. ***Having daughters leads to more left-leaning political preferences.***

(Y for respondents with daughters) > (Y for respondents without daughters).

This study investigates whether having daughters influences individuals' left-leaning political preferences. The analysis will rely on survey data from the 2005, 2009, 2014 and 2019 waves of the SOEP.

Variables

Outcome variable: political preference

- The dependent variable is a measure of left-right political orientation, typically captured in SOEP by a self-placement on a 0–10 scale, where 0 = far left and 10 = far right. The variable will be reverse-coded so that higher values indicate more left-leaning preferences.

Main independent variable, instrumental variable and control variables are the same as H1.

All variables will be weighted using the appropriate survey weights (phrf) to ensure representativeness.

Inclusion and exclusion criteria

- Survey year: 2005, 2009, 2014, and 2019 (depending on political preference variable availability).
- Age range: individuals from the age 18 and 60 years old will the included in the sample to capture working-age adults.
- Parental status: only individuals with at least one biological child will be included, as the analysis focuses on the effect of having daughters, and to apply the instrument that condition is necessary.
- Respondents with missing or invalid values in key variables (e.g., political preferences and child sex) will be excluded from the final analysis.

Exact analysis

To test the hypothesis that having daughters leads to le left-leaning political preferences, I will estimate a pooled two-stage least squares (2SLS) instrumental variable regression. This approach is chosen to identify a causal relationship while addressing potential endogeneity in the main independent variable. An instrument should be a variable that affects whether someone has a daughter but does not directly affect gender attitudes. The variable chosen in the sex of the first child since whether the first child is a girl is random. The strength and validity of the instrument will be assessed.

Given that the instrument (first_daughter) is fixed over time, the panel is unbalanced, and only four survey waves are available, I will estimate a pooled 2SLS model with standard errors clustered at the individual level. This allows me to use all available observations while accounting for within-individual correlation over time.

The main regression model is specified as follows:

- Dependent variable: **:** leftleaning – reverse-coded political orientation scale (higher values = more left-leaning).
- Endogenous independent variable: hasdaughter – dummy for having at least one daughter.
- Instrumental variable: first_daughter – dummy for whether the first child is a girl.
- Controls: educ – categorical variable for education level, age, sex and log_hhnetincome – log of household net income.

Interpretation of results

To evaluate the hypothesis, I will focus on the estimated coefficient of the variable hasdaughter in the second stage of the instrumental variable regression. The coefficient indicates the direction and magnitude of the effect and the p-value assess statistical significance.

If the coefficient on hasdaughter is positive and statistically significant, this will be interpreted as supporting the hypothesis that having daughters leads to left-leaning political preferences.

If the coefficient is not statistically significant, or if it is negative, the result will be interpreted as not supporting the hypothesis.

H3. ***Having daughters leads to more gender equality in the household (measured by behavior).***

(Z for respondents with daughters) > (Z for respondents without daughters).

This study investigates whether having daughters influences couples' gender equality in the household measured by behaviors. The analysis will rely on survey data from the 2000 to 2020 waves of the SOEP.

Variables

Outcome variable: female share of household work

The dependent variable is the female share of total household work, calculated as the proportion of total household hours spent on domestic tasks that are performed by women in workdays (since few respondents answered about the weekend activities). The time-use variables to be used are:

- pli0044_h → hchildcare (childcare)
- pli0043_h → hhousework (housework)
- pli0040 → herrands (errands)
- pli0049_h → hrepairs (repairs)

The total individual housework is calculated as the sum of these four components and the couples level totals are computed by summing across the partners. Then the female share is calculated.

Main independent variable, instrumental variable and control variables are the same as H1.

All variables will be weighted using the appropriate survey weights (phrf) to ensure representativeness.

Inclusion and exclusion criteria

- Survey years: 2000 to 2020.
- Age range: individuals from the age 18 and 60 years old will the included in the sample to capture working-age adults.
- Parental status: only individuals with at least one biological child will be included, as the analysis focuses on the effect of having daughters, and to apply the instrument that condition is necessary.
- Partners: the gender equality behavior in the household is measured among couples.
- Respondents with missing or invalid values in key variables (e.g., household work and child sex) will be excluded from the final analysis.

Exact analysis

To test the hypothesis that having daughters leads to more gender equality in the household, I will estimate a pooled two-stage least squares (2SLS) regression model. This approach is designed to identify a causal relationship between having daughters and the share of household work performed by women, while addressing potential endogeneity in the main independent variable. An instrument should be a variable that affects whether someone has a daughter but does not directly affect gender attitudes. The variable chosen in the sex of the first child since whether the first child is a girl is random. The strength and validity of the instrument will be assessed.

Given that the instrument (first_daughter) is fixed over time, I will estimate a pooled 2SLS model with standard errors clustered at the individual level. This allows me to use all available observations while accounting for within-individual correlation over time, including year fixed effects to control for time trends

The main regression model is specified as follows:

- Dependent variable: female_hwork_share (higher values = less equality).
- Endogenous independent variable: hasdaughter – dummy for having at least one daughter.
- Instrumental variable: first_daughter – dummy for whether the first child is a girl.
- Controls: educ – categorical variable for education level, age, sex and log_hhnetincome – log of household net income.

Interpretation of results

To evaluate the hypothesis, I will focus on the estimated coefficient of the variable hasdaughter in the second stage of the instrumental variable regression. The coefficient indicates the direction and magnitude of the effect and the p-value will assess statistical significance.

If the coefficient on hasdaughter is negative and statistically significant, this will be interpreted as supporting the hypothesis that having daughters leads to more gender equality in the household.

If the coefficient is not statistically significant, or if it is positive, the result will be interpreted as not supporting the hypothesis.

H4. ***Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).***

(X for male respondents with daughters− X for male respondents without daughters) > (X for female respondents with daughters− X for female respondents without daughters)

This study investigates whether the impact of having daughters on gender attitudes differs by the respondent's gender. Specifically, it tests whether men experience a stronger shift toward egalitarian views when they have daughters, compared to women. The analysis will rely on survey data from the 2019 wave of the SOEP.

Variables and Inclusion and exclusion criteria

Same as presented in H1.

Exact analysis

To test the hypothesis that having daughters leads to stronger support for gender equality in men than women, I will estimate a two-stage least squares (2SLS) instrumental variable regression with a interaction term between sex and hasdaughter. That is important to test the causal inference of the phenomenon. An instrument should be a variable that affects whether someone has a daughter but does not directly affect gender attitudes. The variable chosen in the sex of the first child since whether the first child is a girl is random. The strength and validity of the instrument will be assessed.

To explore whether the effect of having daughters differs by gender, I will include an interaction term between hasdaughter and the respondent's sex (hasdaughter_female). This allows the model to estimate separate effects for men and women.

The main regression model is specified as follows:

- Dependent variable: genderatt – index of gender attitudes.
- Endogenous independent variable: hasdaughter – dummy for having at least one daughter and hasdaughter_female – interaction term equal to 1 for women who have at least one daughter.
- Instrumental variables: first_daughter – dummy for whether the first child is a girl and first_daughter_female – interaction term equal to 1 for women whose first child is a girl.
- Controls: educ – categorical variable for education level, age, sex and log_hhnetincome – log of household net income.

Interpretation of results

To evaluate the hypothesis, I will focus on the estimated coefficient of the interaction term between hasdaughter and sex (hasdaughter_female) in the second stage of the instrumental variable regression. The coefficient indicates the direction and magnitude of the effect and the p-value assess statistical significance.

If the coefficient on hasdaughter_female is negative and statistically significant, this will be interpreted as supporting the hypothesis that having daughters leads to more egalitarian gender attitudes for men than for women.

If the coefficient is not statistically significant, or if it is positive, the result will be interpreted as not supporting the hypothesis.